

Micromagnetic Simulations using mumax³

Danny Thonig

School of Science and Technology, Örebro University, Örebro, Sweden

REFERENCES

- <https://mumax.github.io/index.html>
- A. Vansteenkiste et al. “The design and verification of MuMax3” *AIP Advances* 4, 107133 (2014)
- J Leliaert et al “Fast micromagnetic simulations on GPU—recent advances made with mumax3” *J. Phys. D: Appl. Phys.* **51** 123002 2018
- Jonas J. Joos et al “Tutorial: Simulating modern magnetic material systems in mumax3” *J. Appl. Phys.* 134, 171101 (2023)

Future

<https://neuralmag.org>

MICROMAGNETISM

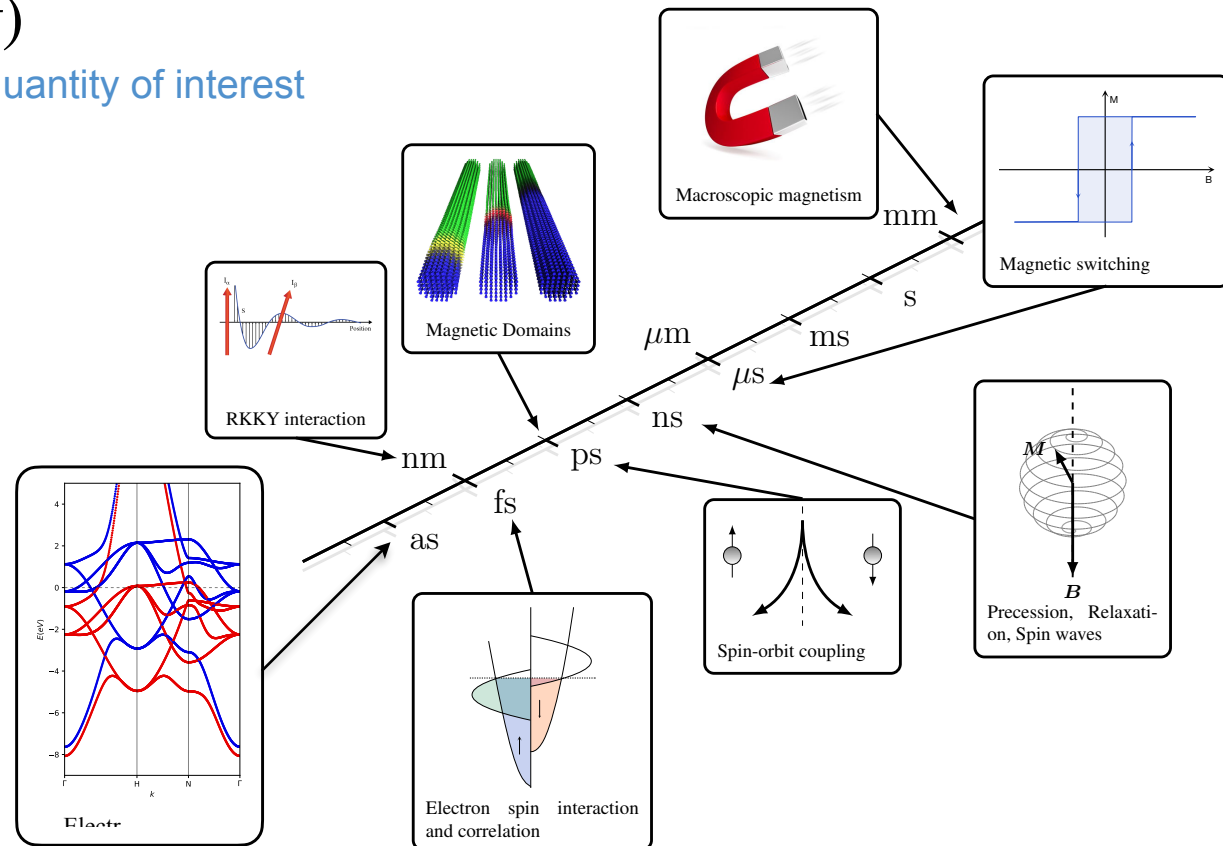
In ferromagnets, neighboring magnetic moments have the tendency to align



Magnetization can be described by a continuous vector field

$$\mathbf{M}(\mathbf{r}, t) = M_s(\mathbf{r}) \mathbf{m}(\mathbf{r}, t)$$

Central quantity of interest



MICROMAGNETISM

The physics is fully described by the total magnetic energy functional:

$$E[\mathbf{m}] = \int_V \left\{ A(\nabla \mathbf{m})^2 - \mu_0 \mathbf{M} \cdot \mathbf{H}_{ext} - \frac{\mu_0}{2} \mathbf{M} \cdot \mathbf{H}_{demag} + \text{other} \right\} d\mathbf{r}$$

Other terms contain e.g.

- Crystal anisotropy
- Dzyaloshinskii-Moriya
- Magneto-elasticity
- Higher-order exchange

Two main objectives

Time integration *dynamics*

- Spin waves
- Domain wall motion
- Spin transfer torques
- Vortex excitation
- ...

Energy minimization *statics*

- Stable magnetic states
- Hysteresis curves
- Phase diagrams
- Domain wall profiles
- ...

MICROMAGNETISM - TIME INTEGRATION

Magnetization dynamics via **Landau-Lifshitz-Gilbert equation**

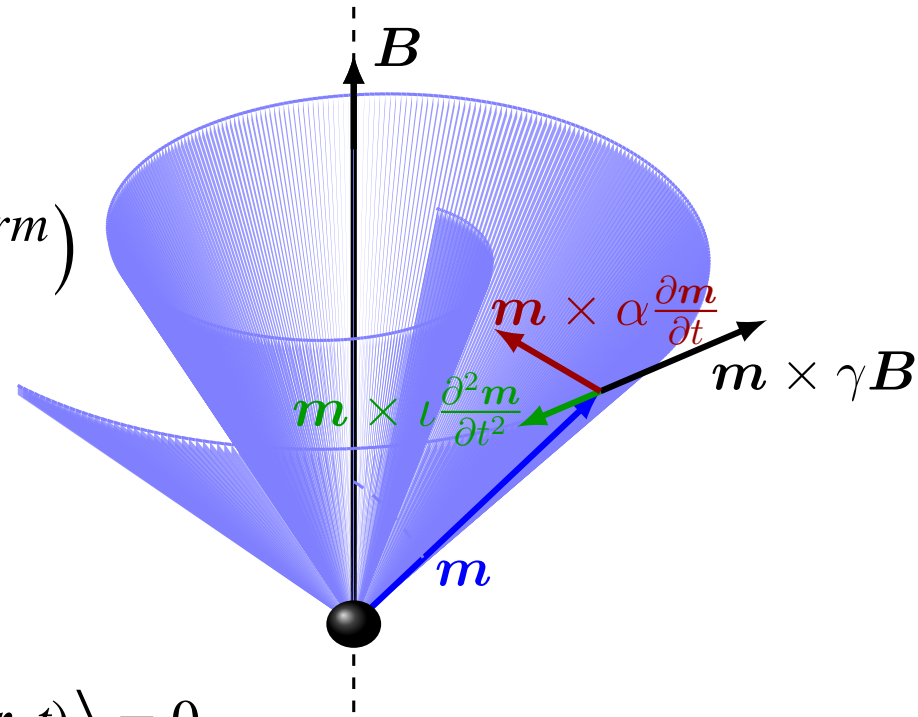
$$\frac{\partial \mathbf{m}}{\partial t} = \mathbf{m} \times \left(-\gamma \mathbf{B} + \mathbf{B}^{diss} + \mathbf{B}^{therm} \right)$$

Intrinsic field $\mathbf{B} = -\frac{1}{\mu_0 M_s} \frac{\partial E}{\partial \mathbf{m}}$

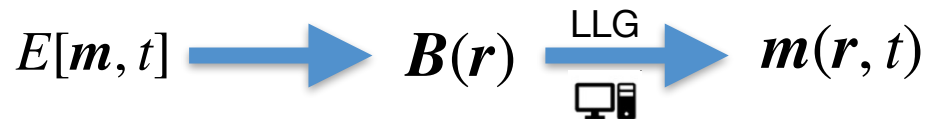
Dissipation field $\mathbf{B}^{diss} = \frac{\alpha}{M_s} \frac{\partial \mathbf{m}}{\partial t}$

Thermal field $\mathbf{B}^{therm}$ with $\langle \mathbf{B}^{therm}(\mathbf{r}, t) \rangle = 0$

$$\langle \mathbf{B}^{therm}(\mathbf{r}, t) \mathbf{B}^{therm}(\mathbf{r}', t') \rangle = 2\alpha \frac{k_B T}{M_s \gamma} \delta(\mathbf{r} - \mathbf{r}') \delta(t - t')$$



Computation of the magnetization dynamics:



MICROMAGNETISM - ENERGY MINIMIZATION

Two approaches to minimize the free energy functional numerically:

1. Standard minimization scheme (*e.g. steepest gradient*)
2. LLG equation with strong damping (*=removing the precession term*)

Continue in google colab and with notebook
mumax3 - ESMA Summer School
2026.ipynb



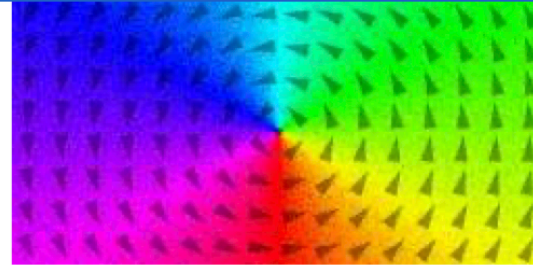
Content

0. Introduction to the code
 1. Phase diagram
 2. Skyrmion Race Track
 3. Hysteresis

```
m = uniform(1,1,0)
```



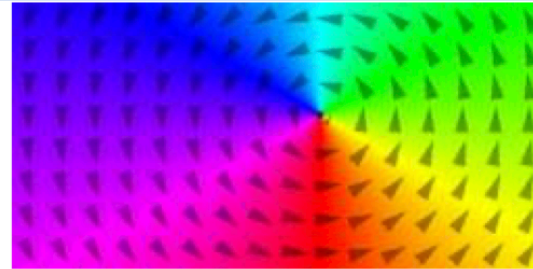
```
m=Vortex(1,-1).Add(0.1,randomMag())
```



```
M = BlochSkyrmion(1, 1)
```



```
m=Vortex(1,-1).transl(100e-9,50e-9,0)
```



```
m = uniform(1, 1, 1)  
m.setInShape(cylinder(400e-9,100e-9), vortex(1,-1))
```

